

Vector Fields and Line Integrals

Lecture 04

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Green's Theorem in the Plane

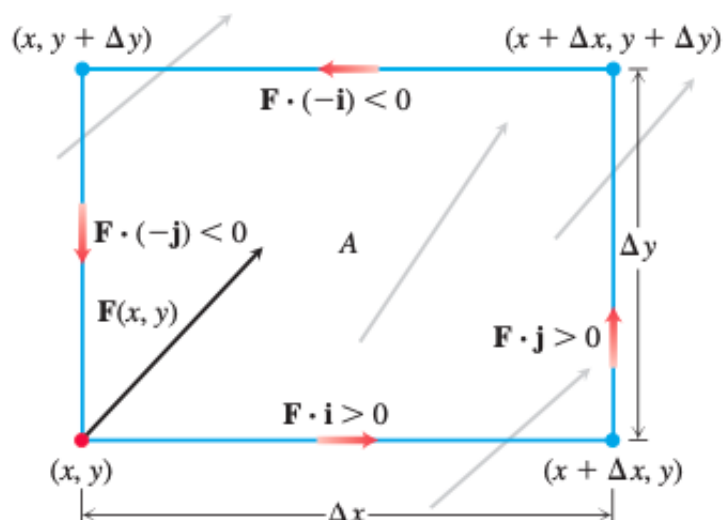
Motivation

- What if $\mathbf{F}$ is not conservative? Can we find the work or flux integral over a closed curve?
- The discussion is given in terms of velocity fields of fluid flows (a fluid is a liquid or a gas) because they are easy to visualize. However, the derivation of Green's theorem applies to any vector field independent of any particular interpretation, provided the underlying assumptions are met.
- Two main ideas will be discussed for Green's Theorem: **circulation density** around an axis perpendicular to the plane and **divergence or flux density**.

Spin around an Axis: The k -th component of Curl

Suppose that $\mathbf{F}(x, y) = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$ is the velocity field of a fluid flowing in the plane and that the first partial derivatives of M and N are continuous at each point of region R .

Construction:



Mathematical treatment:

The rate at which a fluid flows along the bottom edge of a rectangular region A in the direction $\mathbf{i}$ is approximately $\mathbf{F}(x, y) \cdot \mathbf{i}\Delta x$, which is positive for the vector field $\mathbf{F}$.

To approximate the rate of circulation at the point (x, y) , we calculate the (approximate) flow rates along each edge in the directions of the red arrows, sum these rates, and then divide the sum by the area of A .

Taking the limit as $\Delta x \rightarrow 0$ and $\Delta y \rightarrow 0$ gives the rate of the

circulation per unit area.

$$\text{Top} \rightarrow \mathbf{F}(x, y + \Delta y) \cdot (-\mathbf{i})\Delta x = -M(x, y + \Delta y)\Delta x$$

$$\text{Bottom} \rightarrow \mathbf{F}(x, y) \cdot (\mathbf{i})\Delta x = M(x, y)\Delta x$$

$$\text{Right} \rightarrow \mathbf{F}(x + \Delta x, y) \cdot (\mathbf{j})\Delta y = N(x + \Delta x, y)\Delta y$$

$$\text{Left} \rightarrow \mathbf{F}(x, y) \cdot (-\mathbf{j})\Delta y = -N(x, y)\Delta y$$

Summing up in pairs,

$$\text{Top and bottom} \rightarrow -[M(x, y + \Delta y) - M(x, y)]\Delta x$$

$$\text{Right and Left} \rightarrow [N(x + \Delta x, y) - N(x, y)]\Delta y$$

$$\text{Top and bottom} \approx -\left(\frac{\partial M}{\partial y}\Delta y\right)\Delta x$$

$$\text{Right and Left} \approx \left(\frac{\partial N}{\partial x}\Delta x\right)\Delta y$$

Adding these two equations gives the net circulation rate relative to the counterclockwise direction

$$\text{Circulation rate around rectangle} \approx \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)\Delta x\Delta y$$

We now divide by $\Delta x\Delta y$ to get the circulation rate per unit area or **circulation density for the rectangle**

$$\frac{\text{Circulation rate around rectangle}}{\text{rectangle area}} \approx \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)$$

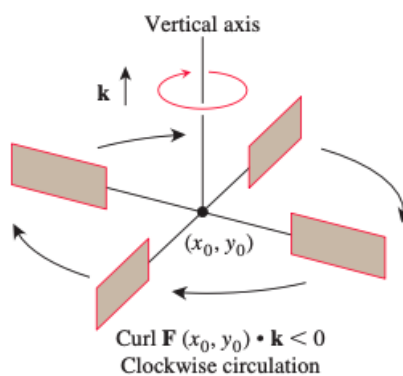
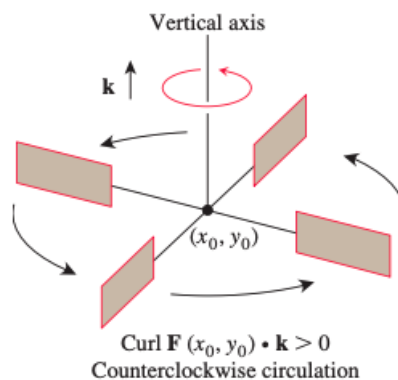
Definition

The **circulation density** of a vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ at the point (x, y) is the scalar expression

$$\text{Circulation density} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

This expression is also called the **k th-component of the curl**, defined by $(\text{curl}\mathbf{F}) \cdot \mathbf{k}$.

Graphically,



In the flow of an incompressible fluid over a plane region, the **k th-component of the curl measures the rate of the fluid's rotation at a point**. The k th-component of the curl is **positive** at points where the rotation is **counterclockwise** and **negative** where the rotation is **clockwise**.

Example

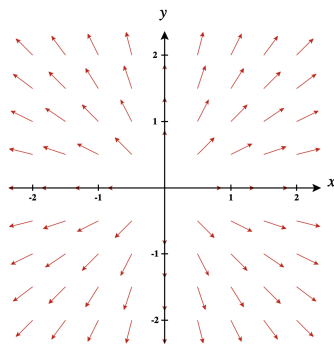
The following vector fields represent the velocity of a gas flowing in the xy -plane. Find the circulation density of each vector field and interpret its physical meaning.

1. Uniform expansion or compression: $\mathbf{F}(x, y) = cx\mathbf{i} + cy\mathbf{j}$
2. Uniform rotation: $\mathbf{F}(x, y) = -cy\mathbf{i} + cx\mathbf{j}$
3. Shearing flow: $\mathbf{F}(x, y) = y\mathbf{i}$
4. Whirlpool effect: $\mathbf{F}(x, y) = -\frac{y}{x^2 + y^2}\mathbf{i} + \frac{x}{x^2 + y^2}\mathbf{j}$

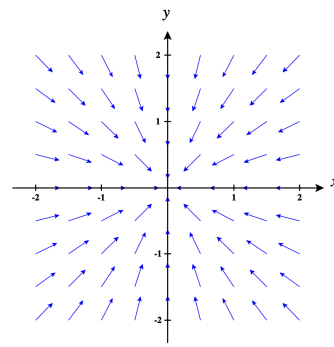
1. Uniform expansion or compression:

$$\text{curl } \mathbf{F} \cdot \mathbf{k} = \frac{\partial}{\partial x}(cy) - \frac{\partial}{\partial y}(cx) = 0$$

This means that the gas is not circulating at a very small scales. The gas is infact exapnding.



(a) $c > 0$

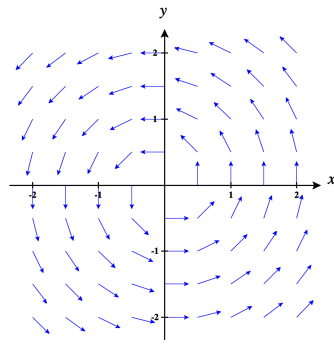


(b) $c < 0$

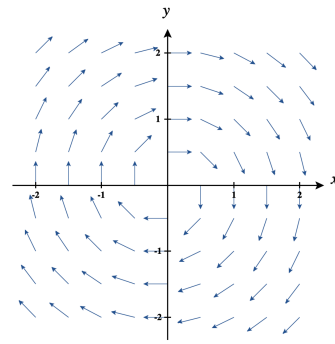
2. Rotation:

$$\text{curl } \mathbf{F} \cdot \mathbf{k} = \frac{\partial}{\partial x}(cx) - \frac{\partial}{\partial y}(-cy) = 2c$$

The constant circulation density indicates rotation at every point. If $c > 0$, the rotation is counterclockwise; if $c < 0$, the rotation is clockwise.



(c) $c > 0$

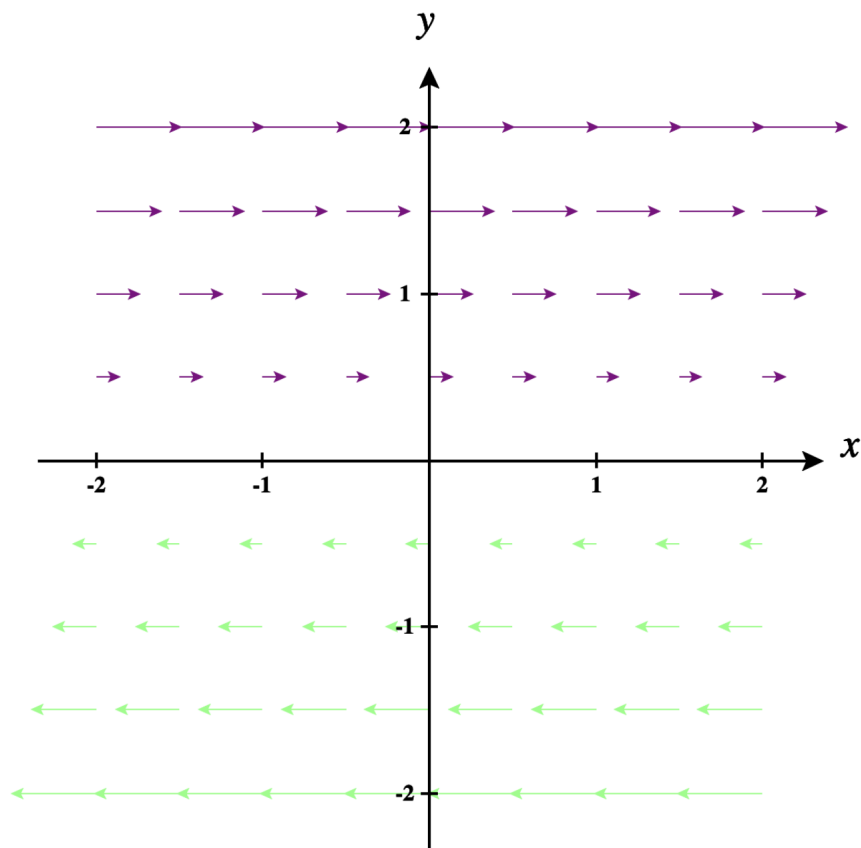


(d) $c < 0$

3. Shear:

$$\text{curl } \mathbf{F} \cdot \mathbf{k} = -\frac{\partial}{\partial y}(y) = -1$$

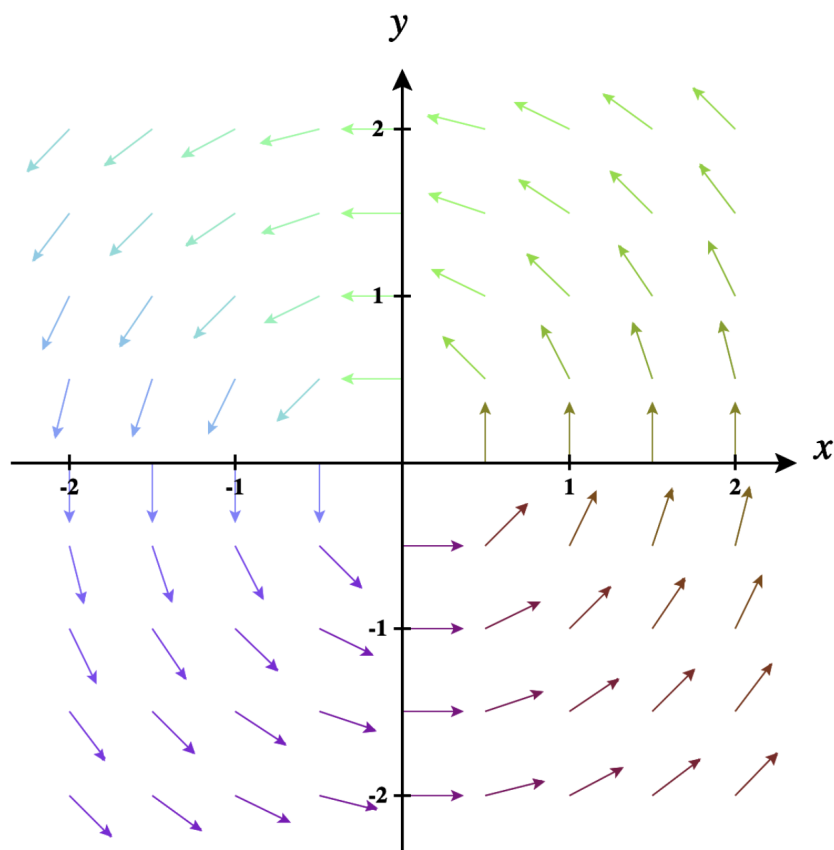
Since the circulation density is uniform and negative, a paddle wheel placed in the fluid will rotate clockwise under the shearing flow. The rate of rotation is same at each point. The average effect of the fluid flow is to push fluid clockwise.



4. Whirlpool:

$$\text{curl } \mathbf{F} \cdot \mathbf{k} = \frac{\partial}{\partial x} \left(\frac{x}{x^2 + y^2} \right) - \frac{\partial}{\partial y} \left(\frac{-y}{x^2 + y^2} \right) = 0$$

The circulation density is 0 at every point away from the origin (where the vector field is undefined and the whirlpool effect is taking place). The gas is not circulating at any point for which the vector field is defined.



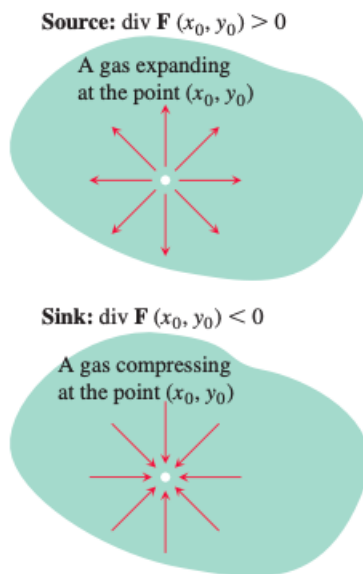
Remark: Although the field looks like it is rotating, the curl is zero everywhere away from the origin, so a small paddle wheel won't spin locally. The apparent whirlpool effect comes from the singularity at the origin, which creates nonzero circulation around closed loops.

Divergence

The **divergence (flux density)** of a vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ at the point (x, y) is

$$\text{Flux density} = \text{div } \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}$$

Graphically,

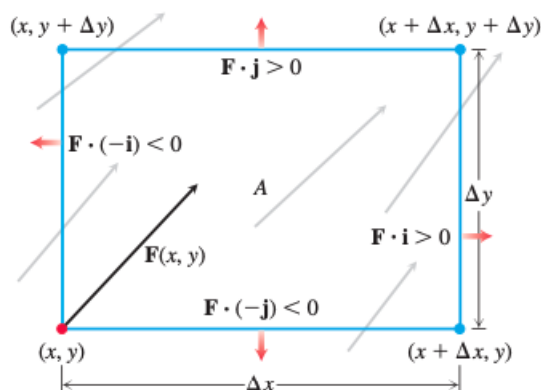


If a gas is expanding at a point (x_0, y_0) , then the lines of flow have positive divergence; if the gas is compressing the divergence is negative.

Inquiry Question

How to prove the Flux density?

Hint: To prove, consider the mechanism



The rate at which the fluid leaves the rectangular region A across the bottom edge in the direction of outward normal $-\mathbf{j}$ is approximately $\mathbf{F}(x, y) \cdot (-\mathbf{j})\Delta x$. Follow the same construction to lead to the final conclusion.

Example

The following vector fields represent the velocity of a gas flowing in the xy -plane. Find the divergence, and interpret what it means for each vector field.

1. Uniform expansion or compression: $\mathbf{F}(x, y) = cx\mathbf{i} + cy\mathbf{j}$
2. Uniform rotation: $\mathbf{F}(x, y) = -cy\mathbf{i} + cx\mathbf{j}$
3. Shearing flow: $\mathbf{F}(x, y) = y\mathbf{i}$
4. Whirlpool effect: $\mathbf{F}(x, y) = -\frac{y}{x^2 + y^2}\mathbf{i} + \frac{x}{x^2 + y^2}\mathbf{j}$